

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No:- 22	
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Class	CSE A2
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Title of Assignment:	Demonstration and exploration of JDBC classes and interfaces (PreparedStatement, and ResultSet).
Practice questions **Output Screenshots Solve questions on given link and attach screen shot of leetcode with your login and o/p**	1. Create a login application using PreparedStatement to check username and password. import java.sql.Connection; import java.sql.DriverManager; import java.sql.PreparedStatement; import java.sql.ResultSet; import java.util.Scanner; public class HospitalStaffLogin { private static final String DB_URL = "jdbc:mysql://localhost:3306/hospital_db"; private static final String DB_USER = "root"; private static final String DB_PASSWORD = "arpit@2006";

```
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("=== Hospital Staff Portal ===");
        System.out.print("Enter Staff ID: ");
        String staffId = scanner.nextLine();

        System.out.print("Enter Password: ");
        String password = scanner.nextLine();



        String query = "SELECT role FROM hospital_staff WHERE staff_id = ? AND
password = ?";

        try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER,
DB_PASSWORD);
             PreparedStatement pstmt = conn.prepareStatement(query)) {

            pstmt.setString(1, staffId);
            pstmt.setString(2, password);

            try (ResultSet rs = pstmt.executeQuery()) {
                if (rs.next()) {
                    String role = rs.getString("role");
                    System.out.println("\n Authentication Successful!");



                    if ("Doctor".equalsIgnoreCase(role)) {
                        System.out.println("Access Granted: Welcome, Dr. " + staffId + ".
Redirecting to Doctor Dashboard (Patient Records & Prescriptions).");
                    } else if ("Nurse".equalsIgnoreCase(role)) {
                        System.out.println("Access Granted: Welcome, Nurse " + staffId + ".
Redirecting to Nursing Station (Vitals & Ward Management).");
```

```
                } else {
                    System.out.println("Access Granted: Welcome, Staff member.");
                }
            } else {
                System.out.println("\n Authentication Failed: Invalid Staff ID or
Password.");
            }
        }

    } catch (Exception e) {
        System.err.println("Database Error: " + e.getMessage());
    }
    scanner.close();
  }
}
```

2. Create a Java application using PreparedStatement to authenticate hospital staff login (doctor/nurse/login ID and password). Display appropriate access messages based on authentication result.

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;

public class HospitalUI extends JFrame {
   private static final String DB_URL = "jdbc:mysql://localhost:3306/hospital_db";
   private static final String DB_USER = "root";
   private static final String DB_PASSWORD = "arpit@2006";
```

```
    public HospitalUI() {
        setTitle("Hospital Portal System");
        setSize(450, 350);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        JTabbedPane tabbedPane = new JTabbedPane();
        tabbedPane.addTab("General Login", createGeneralLoginPanel());
        tabbedPane.addTab("Staff Login", createStaffLoginPanel());

        add(tabbedPane);
    }

    private JPanel createGeneralLoginPanel() {
        JPanel panel = new JPanel(new GridBagLayout());
        GridBagConstraints gbc = new GridBagConstraints();
        gbc.insets = new Insets(8, 8, 8, 8);
        gbc.fill = GridBagConstraints.HORIZONTAL;

        JLabel lblTitle = new JLabel("=== System Login ===",
SwingConstants.CENTER);
        lblTitle.setFont(new Font("Arial", Font.BOLD, 16));

        JLabel lblUser = new JLabel("Username:");
        JTextField txtUser = new JTextField(15);
        JLabel lblPass = new JLabel("Password:");
        JPasswordField txtPass = new JPasswordField(15);
        JButton btnLogin = new JButton("Login");
        JLabel lblStatus = new JLabel("", SwingConstants.CENTER);

        gbc.gridx = 0; gbc.gridy = 0; gbc.gridwidth = 2;
        panel.add(lblTitle, gbc);
```

```
        gbc.gridwidth = 1;
        gbc.gridx = 0; gbc.gridy = 1;
        panel.add(lblUser, gbc);
        gbc.gridx = 1; gbc.gridy = 1;
        panel.add(txtUser, gbc);

        gbc.gridx = 0; gbc.gridy = 2;
        panel.add(lblPass, gbc);
        gbc.gridx = 1; gbc.gridy = 2;
        panel.add(txtPass, gbc);

        gbc.gridx = 0; gbc.gridy = 3; gbc.gridwidth = 2;
        panel.add(btnLogin, gbc);

        gbc.gridx = 0; gbc.gridy = 4; gbc.gridwidth = 2;
        panel.add(lblStatus, gbc);

        btnLogin.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                String username = txtUser.getText();
                String password = new String(txtPass.getPassword());

                String query = "SELECT * FROM users WHERE username = ? AND
password = ?";
                try (Connection conn = DriverManager.getConnection(DB_URL,
DB_USER, DB_PASSWORD);
                     PreparedStatement pstmt = conn.prepareStatement(query)) {

                    pstmt.setString(1, username);
                    pstmt.setString(2, password);

                    try (ResultSet rs = pstmt.executeQuery()) {
                        if (rs.next()) {
```

```
                    lblStatus.setForeground(Color.GREEN.darker());
                    lblStatus.setText("Login Successful! Welcome, " +
rs.getString("username") + ".");
                } else {
                    lblStatus.setForeground(Color.RED);
                    lblStatus.setText("Invalid Username or Password.");
                }
            }
        } catch (Exception ex) {
            lblStatus.setForeground(Color.RED);
            lblStatus.setText("Database Error: " + ex.getMessage());
        }
    }
});

    return panel;
}

private JPanel createStaffLoginPanel() {
    JPanel panel = new JPanel(new GridBagLayout());
    GridBagConstraints gbc = new GridBagConstraints();
    gbc.insets = new Insets(8, 8, 8, 8);
    gbc.fill = GridBagConstraints.HORIZONTAL;

    JLabel lblTitle = new JLabel("=== Hospital Staff Portal ===",
SwingConstants.CENTER);
    lblTitle.setFont(new Font("Arial", Font.BOLD, 16));

    JLabel lblStaffId = new JLabel("Staff ID:");
    JTextField txtStaffId = new JTextField(15);
    JLabel lblPass = new JLabel("Password:");
    JPasswordField txtPass = new JPasswordField(15);
    JButton btnLogin = new JButton("Login");
    JLabel lblStatus = new JLabel("", SwingConstants.CENTER);
```

```
        gbc.gridx = 0; gbc.gridy = 0; gbc.gridwidth = 2;
        panel.add(lblTitle, gbc);

        gbc.gridwidth = 1;
        gbc.gridx = 0; gbc.gridy = 1;
        panel.add(lblStaffId, gbc);
        gbc.gridx = 1; gbc.gridy = 1;
        panel.add(txtStaffId, gbc);

        gbc.gridx = 0; gbc.gridy = 2;
        panel.add(lblPass, gbc);
        gbc.gridx = 1; gbc.gridy = 2;
        panel.add(txtPass, gbc);

        gbc.gridx = 0; gbc.gridy = 3; gbc.gridwidth = 2;
        panel.add(btnLogin, gbc);

        gbc.gridx = 0; gbc.gridy = 4; gbc.gridwidth = 2;
        panel.add(lblStatus, gbc);

        btnLogin.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                String staffId = txtStaffId.getText();
                String password = new String(txtPass.getPassword());

                String query = "SELECT role FROM hospital_staff WHERE staff_id = ?
AND password = ?";
                try (Connection conn = DriverManager.getConnection(DB_URL,
DB_USER, DB_PASSWORD);
                     PreparedStatement pstmt = conn.prepareStatement(query)) {

                    pstmt.setString(1, staffId);
```

```
                pstmt.setString(2, password);

                try (ResultSet rs = pstmt.executeQuery()) {
                    if (rs.next()) {
                        String role = rs.getString("role");
                        lblStatus.setForeground(Color.GREEN.darker());

                        if ("Doctor".equalsIgnoreCase(role)) {
                            lblStatus.setText("Access Granted: Welcome, Dr. " + staffId);
                        } else if ("Nurse".equalsIgnoreCase(role)) {
                            lblStatus.setText("Access Granted: Welcome, Nurse " +
staffId);
                        } else {
                            lblStatus.setText("Access Granted: Welcome, Staff member.");
                        }
                    } else {
                        lblStatus.setForeground(Color.RED);
                        lblStatus.setText("Authentication Failed: Invalid Staff ID or
Password.");
                    }
                } catch (Exception ex) {
                    lblStatus.setForeground(Color.RED);
                    lblStatus.setText("Database Error: " + ex.getMessage());
                }
            }
        });

        return panel;
    }

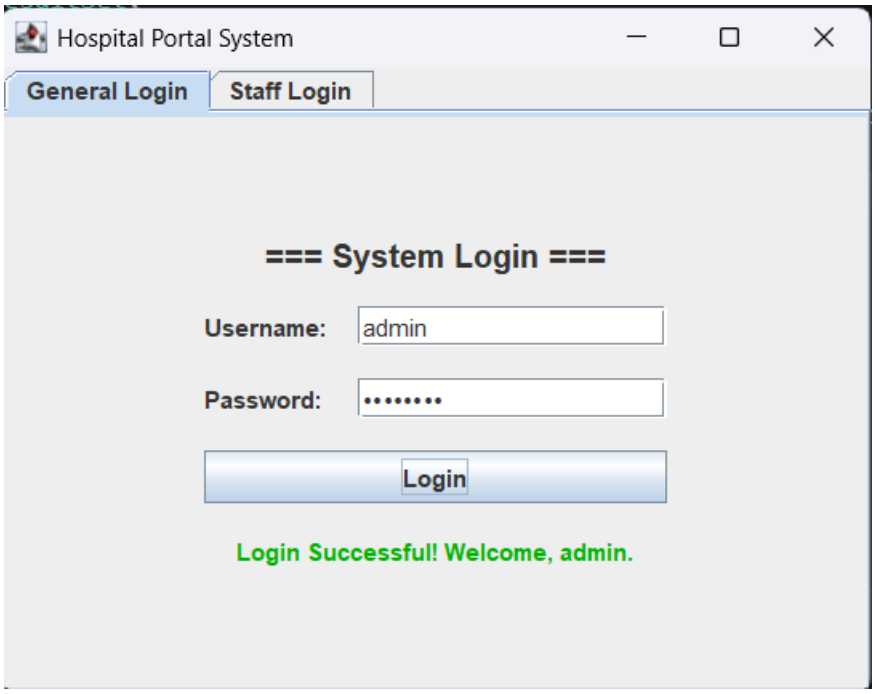
    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> {
            new HospitalUI().setVisible(true);
```

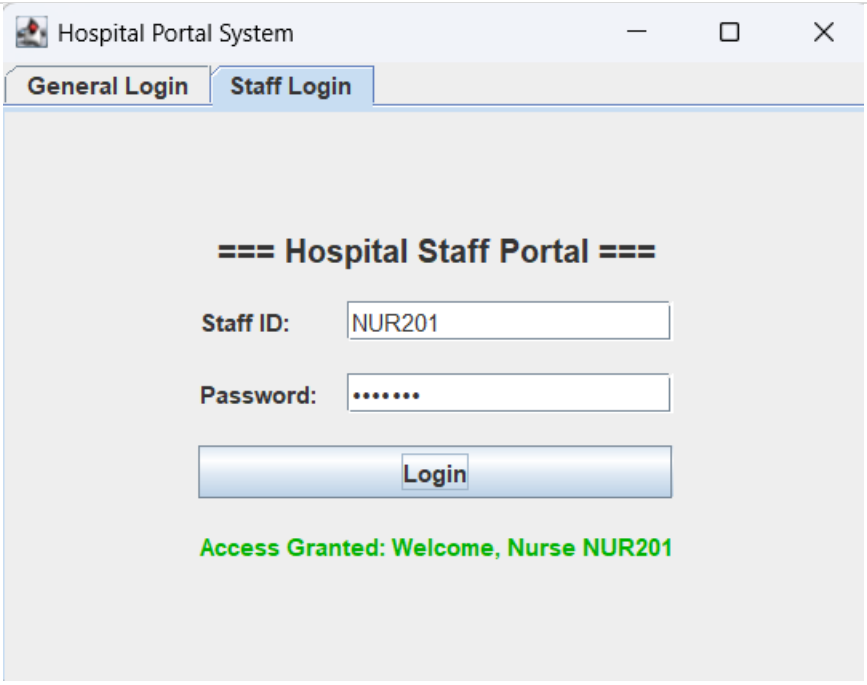
```
        });
    }
}
```

Output Screenshots (if applicable)

```
PS C:\Users\Arpit\OneDrive\Desktop\JDBC_project>  & 'c:\Users\Arpit\.vscode\extensions\redhat.java-1.58.0-win32-x64\jre\25.0.4.1-win32-x86_64\bin\java.exe' '@C:\Users\Arpit\AppData\Local\Temp\cp_4a4cvajz19gmhlalggj62y0q0.argfile' 'HospitalStaffLogin'
=== Hospital Staff Portal ===
Enter Staff ID: DOC101
Enter Password: docpass

 Authentication Successful!
Access Granted: Welcome, Dr. DOC101. Redirecting to Doctor Dashboard (Patient Records & Prescriptions).
```





Conclusion

Conclusion:

Thus, the JDBC classes and interfaces, particularly PreparedStatement and ResultSet, were successfully demonstrated and explored. PreparedStatement is used to execute parameterized SQL queries securely and efficiently, while ResultSet is used to retrieve and process data returned by SQL queries. This practical helped in understanding how Java interacts with a database using JDBC.